Physics Letters B 265 (1991) 182–184
North-Holland

PHYSICS LETTERS B

The analytic value of the light–light vertex graph contributions to the electron $g-2$ in QED

S. Laporta and E. Remiddi

Dipartimento di Fisica, Università di Bologna, and INFN, Sezione di Bologna, I-40126 Bologna, Italy

Received 3 May 1991

The contribution to the $g-2$ of the electron from the light–light graphs at sixth order (three loops) in QED perturbation theory has been evaluated in closed analytical form. The analytic value is in excellent agreement with the most precise numerical evaluation existing in the literature.

We have evaluated in closed analytical form the contribution $a(\gamma\gamma)$ to the anomalous magnetic moment of the electron at sixth order (three loops) in QED perturbation theory from the so-called light–light graphs shown in fig. 1.

The result is

$$a(\gamma\gamma)=+\tfrac{5}{6}\zeta(5)-\tfrac{5}{18}\pi^2\zeta(3)-\tfrac{41}{540}\pi^4-\tfrac{2}{3}\pi^2\ln^2 2$$
$$+\tfrac{2}{3}\ln^4 2+16a_4-\tfrac{4}{3}\zeta(3)-24\pi^2\ln 2+\tfrac{931}{54}\pi^2+\tfrac{5}{9}\,, \tag{1}$$

where $\zeta(p)$ is the Riemann ζ-function of argument p, $\zeta(p)\equiv\sum_{n=1}^{\infty}1/n^p$, [whose first values are $\zeta(2)=\frac{1}{6}\pi^2$, $\zeta(3)=1.202\,056\,903...$, $\zeta(4)=\frac{1}{90}\pi^4$, $\zeta(5)=1.036\,927\,755...$],

$$\ln 2\equiv\sum_{n=1}^{\infty}\frac{1}{2^n n}=0.693\,147\,180...$$

is the logarithm of 2,

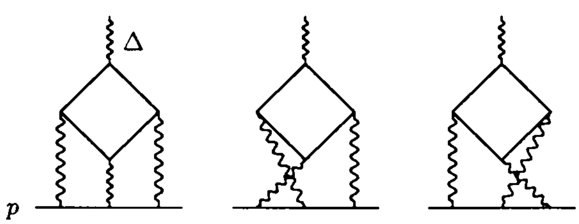

Fig. 1. The light–light graphs (mirror graphs are omitted).

$$a_4\equiv\sum_{n=1}^{\infty}\frac{1}{2^n n^4}=0.517\,479\,061...\,.$$

The numerical value of $a(\gamma\gamma)$ is therefore

$$a(\gamma\gamma)=0.371\,005\,292\,1...\,. \tag{2}$$

The contribution had been previously evaluated only by numerical integration methods; the values in the literature are

$a(\gamma\gamma)=0.36(4)$

(ref. [1]),

$a(\gamma\gamma)=0.37112(8)$

(ref. [2]),

$a(\gamma\gamma)=0.370986(20)$

(ref. [3]),

$a(\gamma\gamma)=0.398(5)$

(ref. [4]).

The results of refs. [1–3] agree within their numerical errors with our result eq. (2), while that of ref. [4] does not.

Due to the perfect agreement with the analytic value (1) of the highly accurate value [3], which is used in the best theoretical value of the electron anomaly a_e of Kinoshita et al. [5], our result does not change in practice the existing theoretical estimate. Indeed, if c_3 is the coefficient of the $(\alpha/\pi)^3$ term in the perturbation expansion of the electron anomaly in QED,

$$a_e(\text{theory}) = \tfrac{1}{2}\alpha/\pi + c_2(\alpha/\pi)^2 + c_3(\alpha/\pi)^3 + c_4(\alpha/\pi)^4 + \dots ,$$

by using eq. (2) we have

$$c_3 = 1.17613(42) , \tag{3}$$

to be compared to the value given in ref. [5]

$$c_3 = 1.17611(42) . \tag{4}$$

The error in eq. (3) is due to the two remaining groups of 15 graphs still not known analytically. We believe that at least one of those groups, the so-called corner-ladder graphs, can be evaluated with a proper extension of the techniques developed for this work. The actual calculation leading to eq. (1) has been quite long; we outline here the main features of the algorithm, leaving a more complete account to a further more exhaustive paper [6].

The main steps of the algorithm are as follows:

(1) The extraction of the scalar contribution to the magnetic moment from the vertex amplitude by the appropriate projection operator. Once the $\Delta \to 0$ limit is taken, for the purpose of the subsequent evaluation the various graphs of fig. 1 can be depicted as in fig. 2, where, in particular, the closed electron loop is drawn as a triangle (of course the Furry theorem does not apply when Δ is ignored). The contribution to $g-2$ is then expressed as sum of 484 different integrals, each term being a polynomial, or rather a monomial, in the loop momenta divided a combination of suitable powers of the denominators appearing in the correspondent self-mass-like scalar Feynman graph (fig. 2). The simplest term, the so-called scalar term, was evaluated in an earlier paper [7].

(2) The use of angular hyperspherical integration

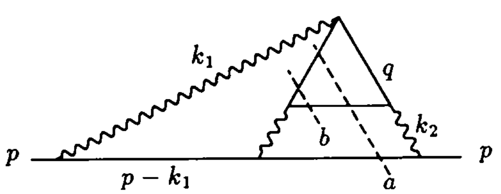

Fig. 2. The self-mass-like scalar graph with the three-body cut of the dispersion relation.

for the k_1 loop combined with a dispersion relation in $(p-k_1)^2$ for the rest of the graph, which is a vertex amplitude depending on p^2, k_1^2 and $(p-k_1)^2$. The discontinuity of the inserted vertex graph consists of a two-body and three-body cut contribution; the three-body cut is shown in fig. 2 by the broken line a. For most of the terms, due to the powers of the loop variables (q, k_2) in the numerators, one must in fact write a suitably subtracted dispersion relation; such subtractions are the main cause of the blowing-up of the size of the calculation. After this stage, each contribution has the form of a triple integral over a combination of dilogarithmic, logarithmic, or rational functions depending on the three variables (l, a, b), where l is the (euclidean) magnitude of the vector k_1^2, a is the integration variable of the dispersion relation in $(p-k_1)^2$, and b is an internal integration variable within the three-body cut for the electron q-loop.

(3) A proper classification scheme and notation for the obtained integrals.

(4) The choice of the order of integration. In the three-body cut four square roots appear:

$$\sqrt{l(l+4)} , \quad \sqrt{b(b-4)} ,$$

$$R(a, -l, 1) , \quad R(a, b, 1) ,$$

where

$$R(x, y, z) \equiv \sqrt{(x-y-z)^2 - 4yz} = \sqrt{x^2+y^2+z^2-2xy-2yz-2xz} .$$

We note that each integration variable appears at least in two square roots. Fortunately the square roots never appear all together, but only three at a time.

If $\sqrt{l(l+4)}$ is missing we integrate first $R(a, -l, 1)$ in l, then $R(a, b, 1)$ on a and b at last; when $R(a, -l, 1)$ is missing we integrate over b first, then on a and l at last. The three-body part must therefore be split into two pieces with a different order of integration, doubling the work. Even worse, these separate contributions are not separately convergent for large l, a, b; it is necessary to parametrize by cutoffs the limits of integration to avoid errors due to improper exchange of limits; of course, the cutoff dependence drops out at the end.

In the two-body cut the square root $R(a, b, 1)$ never appears and a single order of integration suffices (we use a first, b and l at last); the calculation becomes simpler, but in spite of that the amount of the needed algebra is in fact higher.

(5) A number of generalised "integration by parts" formulae for the reduction of the integrals to a standard form.

(6) The direct integration of the integrals brought to the standard from looks discouraging but in fact it is not needed. Given for instance an integral of the form

$$\int_0^\infty \frac{dl}{\sqrt{l(l+4)}} f(l) ,$$

where $f(l)$ is a double integral on the remaining integration variables a and b, one differentiates $f(l)$ with respect to l, evaluates the derivative [which is simpler than the evaluation of $f(l)$ itself] and then obtains $f(l)$ by a quadrature in l.

(7) The procedure is in fact iterated until all the integrals become integrals on a single variable for which a suitable table of definite integrals has been worked out (the actually occurring functions are logarithms, di-, tri- and tetra-logarithms).

All the algebra of all the steps of the calculations was processed relying heavily on the use of the symbolic algebra program ASHMEDAI [8]. We found it convenient to process each partial contribution separately, *one at a time*; this has allowed many consistency checks, as well as approximate but systematic numerical checks of the partial results (most of the separately convergent terms are checked within 1% of precision). The calculations were done on a number of independent, medium-low speed VAXSTATIONS; the complete processing of each of the terms required from a minimum of a couple of hours to a maximum of a few days of continued CPU time (for a total of about 9000 h of CPU time!). We clearly devoted our efforts to devise any possible consistency or cross check rather than to speed up the execution time. As a final check, we evaluated the sum of all the contributions in a single run, which required 5 d of continued CPU time (equivalent to perhaps 14 d of VAX 11/780).

The reliability of ASHMEDAI and its capability of processing large algebraic expressions (with peaks of 10^5 terms or more) was essential for the completion of this calculation. Work at the light–light contribution to the muon anomaly with an electron loop is now in progress.

One of us (E.R.) wish to express his gratitude to M.J. Levine for letting him use his program for algebraic manipulation ASHMEDAI for several years and on a variety of machines; he also thanks the Physics Department of the Carnegie-Mellon University, Pittsburgh, where a key part of the algorithm has been implemented. We also want to thank M. Caffo and S. Turrini for their help in the use of the results of previous work; S. Sorella for a substantial contribution to the three-body cut; A. Mirone, A. Piombo and A. Maurenzi who worked for extended periods at some aspect of the project.

References

[1] J. Aldins, S.J. Brodsky, A. Dufner and T. Kinoshita, Phys. Rev. D 1 (1970) 2378.
[2] T. Kinoshita, Phys. Rev. Lett. 61 (1988) 2898; Phys. Rev. D 40 (1989) 1323.
[3] T. Engelman and M.J. Levine, unpublished, as quoted in: M.J. Levine, H.Y. Park and R.Z. Roskies, Phys. Rev. D 25 (1982) 2205.
[4] M.A. Samuel, Phys. Rev. Lett. 57 (1986) 3133.
[5] T. Kinoshita and W.B. Lindquist, Phys. Rev. D 42 (1990) 636.
[6] S. Laporta, Ph.D. thesis, in preparation;
S. Laporta and E. Remiddi, in preparation.
[7] M. Caffo, E. Remiddi and S. Turrini, Nuovo Cimento A 2 (1984) 220.
[8] M.J. Levine, US AEC Report No. CAR-882-25 (1971), unpublished.